PROBLEMS IN THE NUMERICAL SIMULATION OF MODELS WITH HETEROGENEOUS AGENTS AND ECONOMIC DISTORTIONS

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Abstract
Our work has been concerned with the numerical simulation of dynamic economies with heterogeneous agents and economic distortions. Recent research has drawn attention to inherent difficulties in the computation of competitive equilibria for these economies: A continuous Markovian solution may fail to exist, and some commonly used numerical algorithms may not deliver accurate approximations. We consider a reliable algorithm set forth in Feng et al. (2009), and discuss problems related to the existence and computation of Markovian equilibria, as well as convergence and accuracy properties. We offer new insights into numerical simulation. (JEL: C6, D5, E2)

1. Introduction

In this paper we review some fundamental issues in the computation and numerical simulation of dynamic economic models with heterogeneous agents and market distortions (e.g., externalities, distortionary taxation, and financial frictions). These models are now an integral part of modern macroeconomics and finance, and play a central role in the analysis of fiscal and monetary policies, social security systems and savings over the life cycle, and the determinants of asset price volatility and interest rates.

A main problem with these models is numerical tractability. Economists use numerical simulations to get quantitative predictions. But for models with

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economic distortions or with an infinite number of overlapping generations, their equilibrium solutions cannot be computed by associated global optimization problems as the welfare theorems do not hold. In the presence of multiple competitive equilibria, there is a problem of coordinating expectations that may result in the inability to invoke Bellman's optimality principle.[1] We then have that a Markovian representation of equilibria may only be possible when conditioning over an expanded set of state variables; moreover, as shown in Section 3 this generalized Markovian law of motion may not be a continuous mapping. Therefore, numerical dynamic programming algorithms and other well-known numerical procedures—which search for a continuous policy function—have a limited application for the computation of competitive equilibria of these economies.

These technical issues are mostly ignored in the applied literature, but are crucial to guarantee that a numerical algorithm provides a sufficiently good approximation of the postulated economic model so that we can make reasonable inferences from the output of our computations. Of course, reliable techniques are computationally costly and may not always be feasible, but they should be the starting point for the construction of faster methods with good convergence and accuracy properties.

Examples of non-existence of Markov equilibria can be found in Kubler and Schmedders (2002), Kubler and Polemarchakis (2004), and Santos (2002). Following Duffie et al. (1994), the existence of a Markov equilibrium in a generalized space of variables is proved in Kubler and Schmedders (2003) for an asset pricing model with collateral constraints. Feng et al. (2009) extend these existence results to other economies, and define a Markov equilibrium as a solution over an expanded state of variables that includes the shadow values of investment. The addition of the shadow values of investment as state variables was originally proposed by Kydland and Prescott (1980), and later used in Marcet and Marimon (1998) for recursive contracts, and in Phelan and Stacchetti (2001) for a competitive economy with a representative agent. The main insight of Feng et al. (2009) is to apply this extension of the state space to a reliable algorithm along the lines of Kubler and Schmedders (2003) for the computation of competitive equilibrium models with heterogeneous agents. This redefinition of the state space may be quite effective in the computation of these models. To guarantee convergence to a fixed-point solution the equilibrium operator iterates over a decreasing sequence of candidate compact equilibrium correspondences rather then over functions since no equilibrium function may be continuous. Feng et al. (2009) propose a discretized numerical algorithm that is shown to have good convergence and accuracy properties. This constitutes the first attempt to build numerical foundations

1. For a good discussion of these issues see the early seminal papers of Hellwig (1983) and Kydland and Prescott (1980).

for this type of algorithm in a context in which no equilibrium selection may be continuous.

Our algorithm is also in the spirit of Abreu, Pearce, and Stacchetti (1990) but there are two major differences. First, we will be concerned with developing approximation properties of a discretized algorithm for the simulation of competitive models with heterogeneous agents and market distortions rather than solving for equilibria of super games. Second, their state space includes a vector of continuation utilities, which is not a useful state variable for our purposes.

To illustrate the aforementioned technical difficulties together with potential pitfalls of using non-reliable algorithms, we provide an example in which a regular computational method may actually converge to a wrong fixed-point solution with relatively small Euler equation residuals. In Feng et al. (2009) we consider another example of an overlapping generations economy with chaotic dynamics where our algorithm would work well, but more common algorithms may deliver misleading results. For more complex models with a large number of state variables (e.g., Krusell and Smith 1998) our algorithm may not be computationally feasible, but it is our experience that some other heuristic extensions of standard algorithms may display sizable computational errors.

Finally, we should emphasize that our objective is to compute the set of sequential competitive equilibria, and this set need not be convex. Arbitrary randomizations over the equilibrium correspondence may lead to a drastic expansion of the fixed-point solution. Hence, we compute equilibria directly rather than convex hulls of candidate equilibrium correspondences. Extending the techniques on convex correspondences of Judd, Yeltekin, and Conklin (2003) seems to be a challenging task of rather limited applicability. Our methods could also be applied to the computation of sunspot equilbria. Our operator obtains the set of all equilibria, and therefore it seems possible to compute the set of all sunspot equilibria.

2. The Analytical Framework

Time is discrete, $t = 0, 1, 2, \ldots$. The state of the economy includes a vector of endogenous variables x and vector of exogenous shocks z. Vector x contains all predetermined variables, such as agents' holdings of physical capital, human capital, and financial assets. The exogenous state vector follows a Markov chain $(z_t)_{t\geq 0}$ over a finite set Z. The initial state, $z_0 \in Z$, is known to all agents in the economy. Then $z^t = (z_1, z_2, \ldots, z_t) \in Z^t$ is a history of shocks, often called a date-event or node. Let y denote the vector of all other endogenous variables. These variables could be equilibrium prices or choice variables such as consumption and investment.

Our analysis encompasses models where sequential equilibria exists and can be represented by a sequence $(x_t(z^t), y_t(z^t))_{t=0}^{\infty}$. More important, equilibrium sequences can be characterized as follows. First, the laws of motion for the state variables in vector x are given by a system of non-linear equations:

$$\varphi(x_{t+1}, x_t, y_t, z_t) = 0. \tag{1}$$

Function φ may incorporate technological constraints and individual budget constraints. Let auxiliary variable m be the vector of shadow values of the marginal return to investment for all assets and all agents. This vector is a function of existing variables such as prices, rates of interest, and marginal utilities and productivities:

$$m_t \equiv h(x_t, y_t, z_t). \tag{2}$$

Hence, equilibrium sequences must satisfy the non-linear system of intertemporal optimality, market-clearing, and other restrictions such as short-sales and liquidity requirements[2]

$$\Phi(x_t, y_t, z_t, E_t[m_{t+1}]) = 0, \tag{3}$$

where $E[m]$ is an expectations operator that conditions on state z_t. Finally, it is assumed that φ, h, and Φ are continuous functions, and that vectors m and x lie in a compact domain. Because of this compactness assumption transversality conditions are automatically satisfied.

In quantitative economic analysis, it is common to search for (recursive) equilibria where y_t and x_{t+1} are continuous, time-invariant, functions of x_t and z_t. However, such functions may not exist. We compute sequential equilibria by connecting a sequence of temporary equilibria. A key element of our approach is operator B, which satisfies all the temporary, one–period equilibrium conditions for given future shadow values, $m_+(z_+)$. Furthermore, an iterative procedure based on operator B is guaranteed to converge to the equilibrium correspondence $V^*(x, z)$. This equilibrium correspondence is defined as the set of possible equilibrium values for m, given (x, z). From this fixed-point solution V^*, we can provide a recursive representation of equilibria over an enlarged state space (x, z, m). We formalize these ideas in what follows.

Let z be any initial node, and z_+ be the set of immediate successor states. Then, given (x, z), $B(V)(x, z)$ is the set of all values $m = h(x, y, z)$ with the property that there are current endogenous variables y, and a vector $m_+(z_+) \in V(x_+, z_+)$ that satisfy the temporary equilibrium conditions

$$\begin{aligned} \Phi(x, y, z, E[m_+(z_+)]) &= 0, \\ \varphi(x_+, x, y, z) &= 0. \end{aligned}$$

2. Note that inequalities can always be transformed into equalities by a judicious choice of additional variables.

The following result is proved in Feng et al. (2009):

THEOREM 1. (Convergence) *Let V_0 be a compact-valued correspondence such that $V_0 \supset V^*$. Let $V_n = B(V_{n-1})$, $n \geq 1$. Then, $V_n \to V^*$ as $n \to \infty$. Moreover, V^* is the largest fixed point of the operator B, that is, if $V = B(V)$, then $V \subset V^*$.*

Theorem 1 provides the theoretical foundations for computing equilibria for our class of models. Specifically, this result states that operator B can be applied to any initial guess (correspondence) of possible values $V_0(x, z) \supset V^*(x, z)$ and iterate until a desirable level of convergence to V^* is attained. From operator $B\colon graph(V^*) \to graph(V^*)$ we can select a measurable policy function $y = g^y(x, z, m)$, and a transition function $m_+(z_+) = g^m(x, z, m; z_+)$, for all $z_+ \in Z$. These functions may not be continuous, but give a Markovian characterization of a dynamic equilibrium in the enlarged state space (x, z, m). An important advantage of our approach is that if multiple equilibria exist, we can find all of them. If the equilibrium is always unique, then $B\colon graph(V^*) \to graph(V^*)$ defines a continuous law of motion or Markovian equilibrium over state variables (x, z).

3. A Growth Model with Taxes

The economy is made up of a representative household and a single firm. For a given sequence of interest rates $\{r_t\}$ and taxes $\{\tau_t\}$ the representative household solves the following optimization problem

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t \log(c_t) \\ \text{subject to :} \quad & c_t + k_{t+1} \leq \pi_t + (1 - \tau_t) r_t k_t + T_t. \\ & k_0 \text{ given}, \quad 0 < \beta < 1, \\ & c_t \geq 0, \quad k_{t+1} \geq 0 \text{ for all } t \geq 0. \end{aligned}$$

Here, c_t denotes consumption, k_t is the individual capital holdings. Taxes on capital income $\{\tau_t\}$ are functions of the aggregate capital stock K_t. All tax revenues are rebated back to the consumer as lump-sum transfers T_t.

The representative firm seeks to maximize one-period profits by employing the optimal amount of capital

$$\pi_t = \max_{K_t} \quad f(K_t) - r_t K_t.$$

Let

$$f(K) = K^{1/3}, \quad \beta = 0.95.$$

We then consider the following piecewise linear tax schedule on capital rents:

$$\tau(K) = \begin{cases} 0.10 & \text{if } K \leq 0.160002, \\ 0.05 - 10(K - 0.165002) & \text{if } 0.160002 \leq K \leq 0.170002, \\ 0 & \text{if } K \geq 0.170002. \end{cases}$$

This example is particularly attractive because it can be easily computed by our algorithm for $m = u'(c)f'(k)(1 - \tau(k))$. Santos (2002, Prop. 3.4) shows that a continuous Markov equilibrium fails to exist. For this specification of the model, there are three steady states: The middle steady state is unstable and has two complex eigenvalues while the other two steady states are saddle-path stable. Standard algorithms approximating the Euler equation would solve for a continuous policy function of the form

$$k_{t+1} = g(k_t, \xi),$$

where g belongs to a finite dimensional space of continuous functions as defined by a vector of parameters ξ. We obtain an estimate for ξ by forming a discrete system of Euler equations over as many grid points k^i as the dimensionality of the parameter space:

$$u'(k^i, g(k^i, \xi)) = \beta u'(g(k^i, \xi), g(g(k^i, \xi), \xi)) \cdot [f'(g(k^i, \xi))(1 - \tau(g(k^i, \xi)))].$$

We assume that $g(k^i, \xi)$ belongs to the class of piecewise linear functions, and employ a uniform grid of 5,000 points over the domain $k \in [0.14, 0.19]$. The resulting approximation, together with the highly accurate solution using our algorithm are illustrated in Figure 1.

This approximation of the Euler equation over piecewise continuous functions converged up to computer precision in only three iterations. This fast convergence is actually deceptive because as pointed out previously no continuous policy function does exist. Indeed, the dynamic behavior implied by the continuous function approximation is quite different from the true one as it displays four more steady states, and changes substantially the basins of attraction of the original steady states (see Figure 1). A further test of the fixed-point solution of this algorithm based on the Euler equation residuals produced mixed results. First, the average Euler equation residual (a standard accuracy measure) over the domain of feasible capitals is fairly small (i.e., it is equal to 0.0073). Second, the maximum Euler equation residual is slightly more pronounced in a small area near the unstable steady state. But even in that area, the error is not extremely large: In three tiny intervals the Euler equation residuals are just around 0.06. Therefore, from these computational tests a researcher may be led to conclude that the purported continuous policy function should mimic well the true equilibrium dynamics.

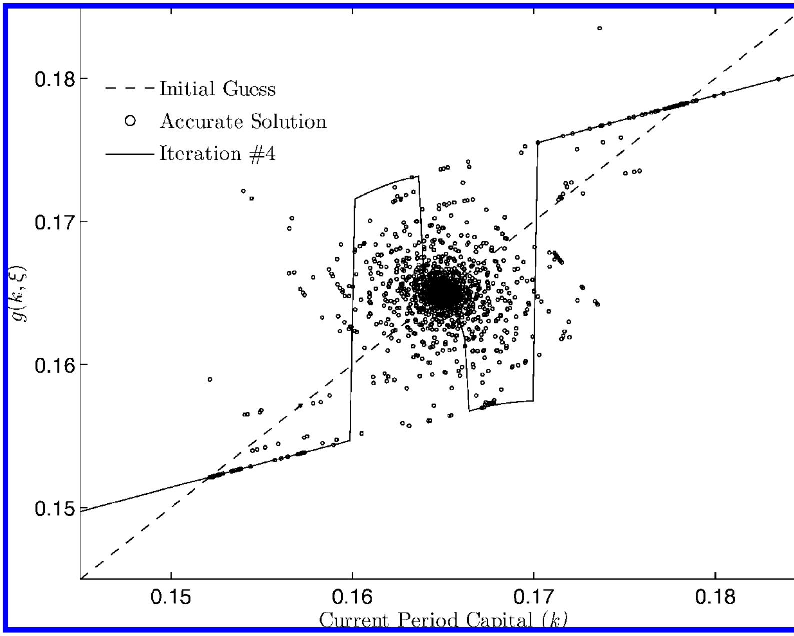

FIGURE 1. Accurate solution vs. continuous policy approximation.

4. Numerical Implementation

We first partition the state space into a finite set of simplices $\{X^j\}$ with non-empty interior and maximum diameter h. Over this partition we define a family of step correspondences that take constant values over each X^j. To obtain a computer representation of a step correspondence we resort to an outer approximation in which each set-value is defined by N elements. Using these two discretizations we obtain a computable approximation of operator B, which we denote by $B^{h,N}$. By a suitable selection of an initial condition V_0 and of these outer approximations, the sequence $\{V_{n+1}^{h,N}\}$ defined recursively as $V_{n+1}^{h,N} = B^{h,N} V_n^{h,N}$ converges to a limit point $V^{*,h,N}$, which must contain the equilibrium correspondence V^*. Again, if the equilibrium is always unique then these approximate solutions would converge uniformly to the continuous Markovian equilibrium law of motion. The following result is proved in Feng et al. (2009):

THEOREM 2. (Accuracy) *For given h, N, and initial condition $V_0 \supseteq V^*$, consider the recursive sequence $\{V_{n+1}^{h,N}\}$ defined as $V_{n+1}^{h,N} = B^{h,N} V_n^{h,N}$. Then, (i) $V_n^{h,N} \supseteq V^*$ for all n; (ii) $V_n^{h,N} \to V^{*,h,N}$ uniformly as $n \to \infty$; and (iii) $V^{*,h,N} \to V^*$ as $h \to 0$ and $N \to \infty$.*

To assess model predictions, analysts usually calculate moments of the simulated paths $(x_t(z^t), y_t(z^t))_{t=0}^{\infty}$ from a numerical approximation. The idea is that the simulated moments should approach those obtained from the original model. Assuming that the optimal policy is a continuous function, Santos and Peralta-Alva (2005) establish various convergence properties of the simulated moments. They also provide examples of non-existence of stochastic steady-state solutions for non-continuous functions, and lack of convergence of empirical distributions to some invariant distribution of the model. We now briefly outline a reliable simulation procedure that circumvents the lack of continuity of the equilibrium law of motion. We are just advancing arguments from our new paper (Santos and Peralta-Alva 2009). We summarize these arguments as follows:

Simulation of the computed equilibrium laws of motion $y = g_n^{y,h,N}(x, z, m)$, *and* $m_+(z_+) = g_n^{m,h,N}(x, z, m; z_+)$. We prove that there are tight upper *USM* and lower *LSM* bounds such that with probability one the corresponding moments from simulated paths $(x_t(z^t), y_t(z^t))_{t=0}^{\infty}$ of these approximate functions stay within the prescribed bounds. More precisely, let $s = (x, y, m)$ and $f : S \times Z \to R_+$ be a function of interest. Let $(\sum_{t=0}^{T} f(s_t, z_t))/T$ represent a simulated moment or some other statistic. Then, with probability one, every limit point of $(\sum_{t=0}^{T} f(s_t, z_t))/T$ must be within the corresponding bounds *LSM* and *USM*.

Existence of an invariant distribution for the original model. An invariant distribution may actually not exist. One way to guarantee existence is to convexify the equilibrium correspondence. Thus, following Blume (1982) and Duffie et al. (1994) we construct a new operator B^{cv} that is a convex-valued correspondence in the space of probability measures. This correspondence has an invariant distribution $\mu^* \in B^{cv}(\mu^*)$.

Accuracy of the simulated moments. For every $\varepsilon > 0$ we can consider a sufficiently good discretized operator $B^{h,N}$ and equilibrium correspondence $V_n^{h,N}$ such that for every simulated path $(s_t, z_t)_{t=0}^{\infty}$ there are invariant distributions μ^*, μ'^* of B^{cv} satisfying $\int f(s, z) d\mu^* - \varepsilon \leq (\sum_{t=0}^{T} f(s_t, z_t))/T \leq \int f(s, z) d\mu'^* + \varepsilon$, for large T, almost surely. Of course, if B^{cv} has a unique invariant distribution μ^* then $\mu'^* = \mu^*$ and the above expression read as

$$\int f(s, z) d\mu^* - \varepsilon \leq \left(\sum_{t=0}^{T} f(s_t, z_t) \right) / T \leq \int f(s, z) d\mu^* + \varepsilon.$$

Duffie et al. (1994) argue that operator B^{cv} allows for some form of sunspot equilibria since the randomization proceeds over equilibrium distributions rather than over an extraneous sunspot variable. This is not, however, the interpretation that we want to give to our simulations. For us, the primitive elements in our analysis are the Markovian equilibrium functions which are obtained from the original equilibrium correspondences without performing arbitrary randomizations. We are therefore computing a numerical approximation of our original model, and approximating the simulated moments generated by our original model. We only consider the invariant distributions of the convex-valued operator B^{cv} to bound the range of variation of the simulated moments for both the approximate and true solutions. These bounds may not be tight when the regularized operator B^{cv} expands substantially the stochastic dynamics.

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